

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

# Chapter 12: Reflection

Step-by-step solutions to the previous year questions

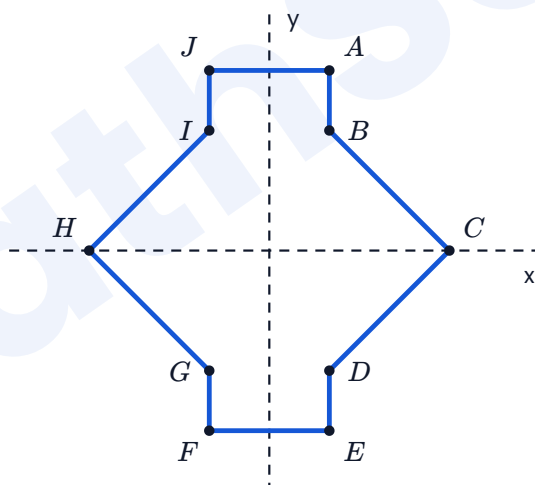
The question numbers match the PYQ question sheet.

## 1 Reflection of figures

1. Plot  $A(1, 3)$ ,  $B(1, 2)$  and  $C(3, 0)$  on graph paper. (a) Reflect  $A$  and  $B$  in the  $x$ -axis to get  $E$  and  $D$ . (b) Reflect  $A$  and  $B$  in the origin to get  $F$  and  $G$ . (c) Reflect  $A$ ,  $B$  and  $C$  in the  $y$ -axis to get  $J$ ,  $I$  and  $H$ . (d) Join  $A, B, C, D, E, F, G, H, I, J$  in order and name the closed figure. **ICSE 2025, 5 marks**

**SOLUTION**

- (a) Reflection in the  $x$ -axis changes the sign of  $y$ :  $E(1, -3)$  and  $D(1, -2)$ .  
 (b) Reflection in the origin changes both signs:  $F(-1, -3)$  and  $G(-1, -2)$ .  
 (c) Reflection in the  $y$ -axis changes the sign of  $x$ :  $J(-1, 3)$ ,  $I(-1, 2)$  and  $H(-3, 0)$ .



- (d) The figure has 10 sides and two of its angles point inwards on each side.

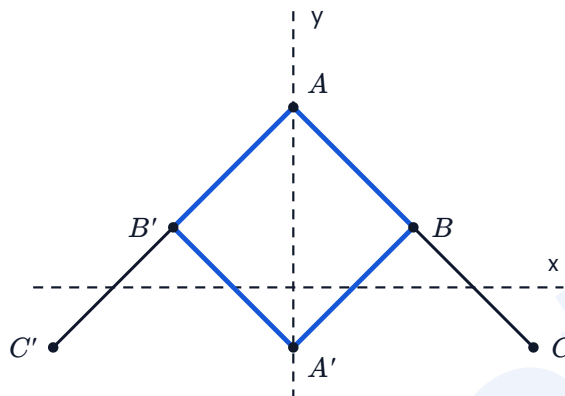
**Answer: a concave decagon, shaped like a double-headed arrow**

2. Plot  $A(0, 3)$ ,  $B(2, 1)$  and  $C(4, -1)$ . (a) Reflect  $B$  and  $C$  in the  $y$ -axis to get  $B'$  and  $C'$ , and write their coordinates. (b) Reflect  $A$  in the line  $BB'$  to get  $A'$ , and write its coordinates. (c) Join  $A, B, A', B'$  in order and name the figure. **ICSE 2024, 5 marks**

**SOLUTION**

(a) Reflection in the  $y$ -axis changes the sign of  $x$ :  $B'(-2, 1)$  and  $C'(-4, -1)$ .

(b)  $BB'$  is the horizontal line  $y = 1$ .  $A$  is 2 units above it, so  $A'$  is 2 units below it:  $A'(0, -1)$ .



(c)  $AB = BA' = A'B' = B'A = \sqrt{2^2 + 2^2} = \sqrt{8}$ , and the diagonals  $AA' = BB' = 4$  are equal.

**Answer:**  $B'(-2, 1)$ ,  $C'(-4, -1)$ ,  $A'(0, -1)$ ;  $ABA'B'$  is a square

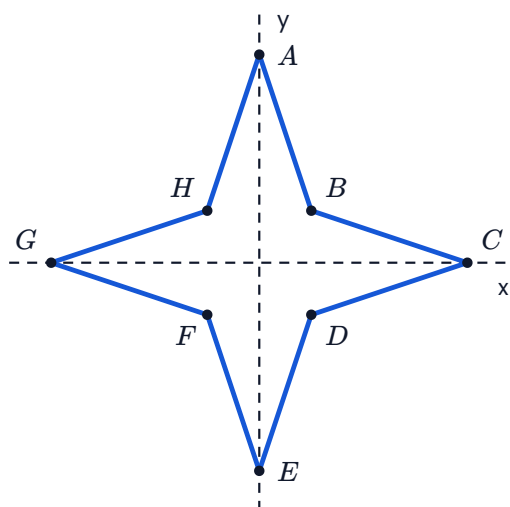
3. Plot  $A(0, 4)$ ,  $B(1, 1)$  and  $C(4, 0)$ . (a) Reflect  $A$  and  $B$  in the  $x$ -axis to get  $E$  and  $D$ . (b) Reflect  $B$  in the origin to get  $F$ , and write its coordinates. (c) Reflect  $B$  and  $C$  in the  $y$ -axis to get  $H$  and  $G$ . (d) Join  $A, B, C, D, E, F, G, H, A$  in order and name the closed figure. **ICSE 2023, 5 marks**

**SOLUTION**

(a)  $E(0, -4)$  and  $D(1, -1)$ .

(b) Reflection in the origin changes both signs:  $F(-1, -1)$ .

(c)  $H(-1, 1)$  and  $G(-4, 0)$ .



(d) The figure has 8 sides, with four of its corners pointing inwards.

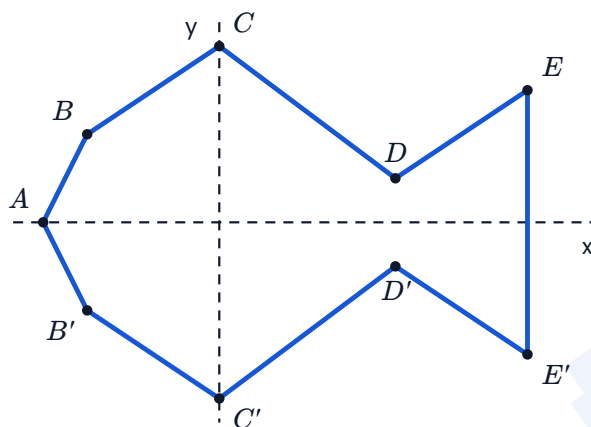
**Answer:**  $F(-1, -1)$ ; the figure is a concave octagon in the shape of a four-pointed star

4. Plot  $A(-4, 0)$ ,  $B(-3, 2)$ ,  $C(0, 4)$ ,  $D(4, 1)$  and  $E(7, 3)$ . Reflect  $B$ ,  $C$ ,  $D$  and  $E$  in the  $x$ -axis to get  $B'$ ,  $C'$ ,  $D'$  and  $E'$ . Join  $A, B, C, D, E, E', D', C', B', A$  in order and name the closed figure.

ICSE 2020, 4 marks

**SOLUTION**

Reflection in the  $x$ -axis changes the sign of  $y$ :  $B'(-3, -2)$ ,  $C'(0, -4)$ ,  $D'(4, -1)$  and  $E'(7, -3)$ .



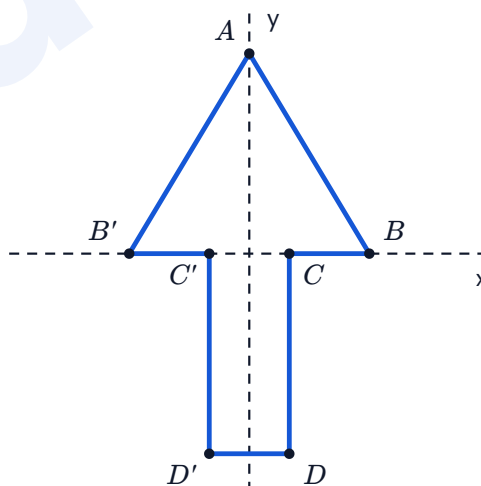
**Answer: the figure is shaped like a fish (a concave nonagon, symmetric about the  $x$ -axis)**

5. Plot  $A(0, 5)$ ,  $B(3, 0)$ ,  $C(1, 0)$  and  $D(1, -5)$ . Reflect  $B$ ,  $C$  and  $D$  in the  $y$ -axis to get  $B'$ ,  $C'$  and  $D'$ , and write their coordinates. Join  $A, B, C, D, D', C', B', A$  in order and name the closed figure.

ICSE 2019, 4 marks

**SOLUTION**

Reflection in the  $y$ -axis changes the sign of  $x$ :  $B'(-3, 0)$ ,  $C'(-1, 0)$  and  $D'(-1, -5)$ .

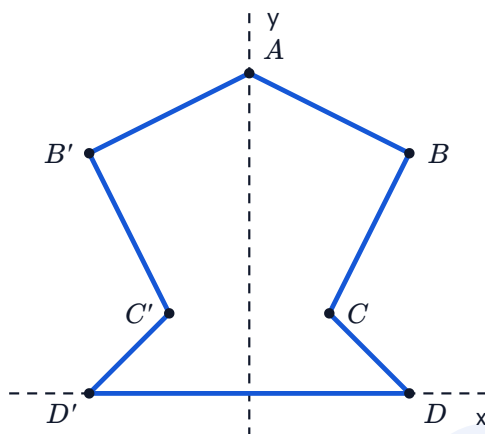


**Answer:  $B'(-3, 0)$ ,  $C'(-1, 0)$ ,  $D'(-1, -5)$ ; the figure is an arrow head (a concave heptagon)**

6. Plot  $A(0, 4)$ ,  $B(2, 3)$ ,  $C(1, 1)$  and  $D(2, 0)$ . Reflect  $B$ ,  $C$  and  $D$  in the  $y$ -axis to get  $B'$ ,  $C'$  and  $D'$ , and write their coordinates. Join  $A, B, C, D, D', C', B', A$  in order and write the equation of the line of symmetry of the figure. **ICSE 2017, 5 marks**

**SOLUTION**

Reflection in the  $y$ -axis changes the sign of  $x$ :  $B'(-2, 3)$ ,  $C'(-1, 1)$  and  $D'(-2, 0)$ .



The figure is its own mirror image in the  $y$ -axis.

**Answer:**  $B'(-2, 3)$ ,  $C'(-1, 1)$ ,  $D'(-2, 0)$ ; the line of symmetry is  $x = 0$

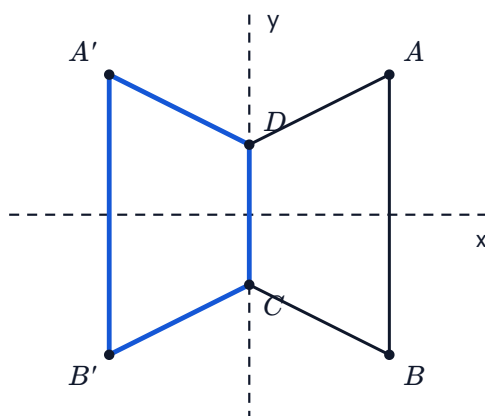
## 2 Invariant points

7.  $ABCD$  is a quadrilateral with vertices  $A(2, 2)$ ,  $B(2, -2)$ ,  $C(0, -1)$  and  $D(0, 1)$ . (a) Reflect  $ABCD$  in the  $y$ -axis to get  $A'B'CD$ , and write the coordinates of  $A'$  and  $B'$ . (b) Name two points that are invariant under this reflection. (c) Name the polygon  $A'B'CD$ . **ICSE 2018, 4 marks**

**SOLUTION**

(a)  $A'(-2, 2)$  and  $B'(-2, -2)$ .

(b)  $C$  and  $D$  lie on the  $y$ -axis, which is the mirror line, so they are invariant.



(c)  $A'B' \parallel DC$  (both vertical) and  $A'D = B'C = \sqrt{5}$ .

**Answer:**  $A'(-2, 2)$ ,  $B'(-2, -2)$ ; invariant points  $C$  and  $D$ ;  $A'B'CD$  is an isosceles trapezium

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